

Inverse or not? How much of B is touched before choosing p

Touched	Method	Cost/iter	Local rate
nothing	steepest descent	$O(n)$	linear, $(\frac{\kappa-1}{\kappa+1})^2$
one product $g^T Bg$	Cauchy point	$O(n^2)$	linear, same rate
j products Bd	CG–Steihaug, Newton–CG	$j O(n^2)$	superlinear*
m recent pairs (s_i, y_i)	L-BFGS	$O(mn)$	linear, fast in practice
rank-2 updated $H_k \approx B^{-1}$	BFGS, DFP, SR1	$O(n^2)$	superlinear
one factorization	Newton, dogleg	$O(n^3)$	quadratic, superlinear
several factorizations	nearly exact TR, mod. Cholesky	$k O(n^3)$	quadratic

No method forms B^{-1} . Down the ladder: more curvature per step, more cost, faster local rate. Chapters 3–7 are its rungs. The bottom rows split again: dense vs. sparse; repair B first (line search) vs. let λ do it (trust region).

*with forcing sequence $\varepsilon_k = \min(\frac{1}{2}, \sqrt{\|g_k\|})\|g_k\|$, Ch. 7